Zhen Wu

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PERSONAL PROFILE

Postdoctoral researcher with a Ph.D. in Geography, specializing in geospatial data science, GIS, and spatiotemporal modeling. Expertise includes:

- Modeling and analyzing urban disasters and climate gentrification under climate change.
- Applying machine learning and big data methods to carbon sink modeling, with strong skills in mapping and data visualization.
- Developing comprehensive indicator systems and quantitative models to assess coastal environmental health.
- Conducting spatiotemporal analyses of land use and land cover change and their socioeconomic impacts using Satellite image data.

EXPERIENCE

Department of Geography and Resource Management, The Chinese University of Hong Kong, Hong Kong SAR, China

Postdoctoral Researcher | August 2025 - Current

My research centers on climate change–induced hazards in coastal regions, using multi-source data to assess and model risks, with a particular focus on climate gentrification.

- Evaluate coastal climate hazards, especially sea-level rise, urban flooding, and extreme heat.
- Develop analytical frameworks for assessing climate gentrification and investigating its driving mechanisms.
- Contribute as a key team member to the preparation of research proposals for competitive funding applications.

School of Geography and Ocean Science, Nanjing University, China

Academic Researcher | July 2024 – July 2025

Modeling and estimating the carbon sinks in China's terrestrial ecosystems using big data techniques, analyzing land use change and its ecological and environmental impacts.

- Estimating carbon sinks in China's terrestrial ecosystems using multi-source remote sensing data and machine learning models.
- Access land use change and carbon dynamics assessment along the Yangtze River.

- Contribute as a key team member to the preparation of research proposals for competitive funding applications.
- As a core member, completed the Guidelines for the Integrative Technology of Ecological Restoration Area Identification and Carbon Sequestration Enhancement Benefits Assessment.

Center for Geographical Analysis (CGA), Harvard University, MA, US

Visiting Scholar | September 2022 – September 2023

Contributed to the development of KNIME, a big-data analytics platform, in the Spatial Data Lab using remote sensing image processing and GIS techniques.

- Played a key role in the Geo-Image development project within the CGA KNIME node team by designing a framework for satellite image processing nodes.
- Taught the theory and applications of Geo-Image in KNIME and supervised hands-on workshop training. Selected teaching experience includes:
  - ✧ Applications of Remote Sensing Data in Carbon Density Estimation, Workshop on Spatiotemporal Innovation, Harvard University (July 2023)
  - ✧ Applications of Geo-AI Models to Remote Sensing Data Using KNIME, Workshop on Spatiotemporal Innovation and Data Modeling, Harbin, China (October 2023)
- Served as a teaching assistant for GIS-related courses, supporting undergraduate students in the use of ArcGIS Pro and QGIS. Courses included Basic Introduction to GIS (Fall 2023), GIS Workshop (Fall 2023), Cartography Workshop Using ArcGIS Pro (Spring 2023), and Harvard GIS Institute (Winter 2022).
- Presented project progress at weekly meetings and major academic or project events.

The Center for International Earth Science Information Network (CIESIN), Columbia University, NY, US

Visiting Scholar | February 2028 – August 2018

Used integrated environmental and ecological indicators to evaluate coastal environmental change and its human impacts.

- Developed a comprehensive marine health index, incorporating environmental, ecological, water quality, fisheries, economic, and human well-being indicators, to evaluate the health of Shanghai's nearshore waters and coastline.
- Assessed the impacts of coastal environmental change on human systems and well-being.

Institute of Scientific Research and Education, Zhejiang University, China

Research and teaching Assistant | July 2019 – July 2020

Worked in a demanding, fast-paced academic environment that required managing multiple responsibilities simultaneously, strong teamwork, and effective communication

with faculty and students. Responsibilities included overseeing graduate academic administration and collaborating with researchers on institutional reports.

- Served as the department's sole teaching assistant, coordinating graduate course schedules, final examinations, and the full academic training process management, including thesis proposals, pre-defense, defense, and graduation. Also managed faculty teaching schedules, handled student administrative affairs, and provided in-class teaching support.
- Supported the Institute of Higher Education by planning research activities, drafting activity proposals, coordinating staffing and schedules for major annual conferences, and preparing conference reports.
- Played a key role in organizing the college's annual gathering for faculty, staff, and students.

School of Geography and Ocean Science, Nanjing University, China

Career Mentor | March 2025 – August 2027

- Provided career guidance to undergraduate and graduate students across the department, including employment advising, presentations at career forums, and individualized recommendations based on students' academic backgrounds and career goals.

Spatial Data Lab (SDL) of Center for Geographic Analysis (CGA), Harvard University, US

Project Mentor | July 2024 – July 2025

- As a project mentor in the Spatial Data Lab, I supervised undergraduate intern Alex Rop in using KNIME to complete the project Drought Index Monitoring Using Multi-Source Remote Sensing Data Based on KNIME Workflows. The project used Landsat imagery to conduct long-term drought analysis in Nairobi, Africa, and Inner Mongolia, China. Its key contribution was the development of a reproducible and transferable workflow that can be adapted for drought monitoring in other regions.

EDUCATION

Doctor of Philosophy, Geography - Nanjing University, 2020-2024

Master of Science, Physical Geography - East China Normal University, 2016 - 2019

Bachelor of Science, Geography Science - Hunan University of Science and Technology, 2011 - 2015

SKILLS/SOFTWARES

GIS & Spatial Analysis: ArcGIS Pro, ArcGIS Online, Arc MAP, QGIS, ENVI, KNIME

Data processing: KNIME, SPSS, Origin

Cloud Platform: Google Earth Engine (GEE)

Programming: Python, JavaScript

PUBLICATIONS

- **Wu, Z.,** Huang, X., Liu, X., et al. **2025**. Land use changes and their effects on poverty in Inner Mongolia, northern China. ***Geography and Sustainability***, 6(4): 100269.
- **Wu, Z.,** Wang, X., Chen, M., et al. **2024**. Changes in sediment N mineralization and immobilization in association with Spartina alterniflora invasion in mangrove wetland. ***Plant and Soil,*** 500: 495–514.
- **Wu, Z.,** Huang, X., Chen, R., et al. **2022.** The United States and China on the paths and policies to carbon neutrality. ***Journal of Environmental Management***, 320: 115785. ***(ESI Highly Cited Paper)***
- **Wu, Z.,** Chen, R., Meadows, M. E., et al. **2021.** Application of the Ocean Health Index to Assess Ecosystem Health for the Coastal Areas of Shanghai, China. ***Ecological Indicators***, 126: 107650.
- **Wu, Z.,** Chen, R., Meadows, M. E., et al. **2019**. Changing Urban Green Spaces in Shanghai: Trends, Drivers and Policy Implications. ***Land Use Policy***, 87: 104080. ***(ESI Highly Cited Paper)***
- **Wu, Z.,** and Chen, R. **2019.** A Comparative Analysis of OHI and PSR Evaluation Methods for Shanghai Ocean Ecosystem Health. ***Journal of East China Normal University***, 4: 174-187.
- **Wu, Z.,** and Huang, X. **2024.** A method for estimating carbon density in terrestrial ecosystems based on multi-source remote sensing data. ***Chinese National Patent (authorized)***, CN118037320A.
- Fu, X., Liu, L. Li, M., Huang, X., **Wu, Z. 2025.** Calibration of 2SFCA and i2SFCA: A Case Study in Shenzhen, China Based on Online Physician Appointment Data. ***Transactions in GIS***. 29, e70061.
- Huang, X., Qi, X., Liu, Z., **Wu, Z.,** et al. **2024.** Optimization of Farmland Spatial Pattern for Carbon Emission Reduction in the Agriculture and Food System. ***Economic Geography***. 44(9), 153-162.
- Wang, Y., Guo, C., ... **Wu, Z. 2021**. Major Contribution to Carbon Neutrality by China's Geosciences and Geological Technologies. ***Journal of China Geology,*** 4: 329-352.

AWARDS

- 2024 Awarded the Second Prize for Outstanding Papers from the Key Laboratory of Carbon Neutrality and Land Spatial Optimization of the Ministry of Natural Resources in China *(Paper title: The United States and China on the paths and policies to carbon neutrality)*
- 2023 Awarded the First Prize of Excellent Scholarship from Nanjing University
- 2022 Awarded the Excellent Student Scholarship from Nanjing University
- 2017 Awarded the Third Prize winner of "The 14th China Graduate Mathematical Modelling Contest" as the team captain

RESEARCH PROJECTS

The Project of Restoration of Degraded Ecosystems and Carbon Sink Dynamics Along the Yangtze River | Jul 2022 – Jul 2025

- Led carbon sink modeling and estimation across the Yangtze River region.
- Conducted spatiotemporal analyses from 1990 onward to identify degraded ecosystems and quantify carbon sink dynamics.
- Generated policy and technical recommendations to inform low-carbon development and ecosystem restoration strategies.

The Research Project of Impacts on Climate Change on Land Use in China and Carbon Neutral Response | Sep 2021 – Jun 2024

- Developed carbon-related models using open-source satellite data to examine land use change and carbon cycle dynamics.
- Assessed spatiotemporal patterns of land use change and their implications for carbon neutrality.

The Project of Ecological Carbon Sink Potential Assessment and Carbon Emission Decrease Path in Shanghai's New City Construction | Jun 2022 – Dec 2022

- Co-authored project proposals and summary reports for carbon sink assessment and emissions reduction planning.
- Produced technical recommendations for local government on carbon sequestration enhancement and low-carbon development pathways.

The Research Project of Spatial Planning in Inner Mongolia's Yellow River Basin | Dec 2020 – Dec 2021

- Modeled land use and ecosystem dynamics under urban expansion scenarios.
- Assessed ecological vulnerability and examined its implications for regional planning and policy.
- Prepared analytical reports to support sustainable spatial planning.

The Research Project of Ocean and Yangtze River Estuary in Shanghai's Ecological Protection | Jan 2017 – Dec 2018

- Processed geospatial datasets and conducted environmental modeling to evaluate coastal ecosystem health.
- Assessed environmental change in coastal and estuarine systems to support ecological protection and sustainable development planning.

PRESENTATIONS

Academic Service

- **Session Organizer and Chair,** "Environmental and Land Resource Challenges in the Context of Climate Change," 2024 Annual Meeting of the American Association of Geographers (AAG), Honolulu, HI, USA, April 2024.
- **Conference Volunteer,** 2024 Annual Meeting of the American Association of Geographers (AAG), Honolulu, HI, USA, April 2024.
- **Conference Volunteer,** 2023 Annual Meeting of the American Association of Geographers (AAG), Denver, CO, USA, March 2023.

Conference Presentations

- "Climate-Induced Hazards and Human Impacts: A Systematic Review." Presented at the European Geosciences Union (EGU) General Assembly 2026, Vienna, Austria, May 2026.
- "Study on the Spatiotemporal Evolution of Carbon Storage in China's Terrestrial Ecosystems." Presented at the World Geography Annual Conference 2024, Shanghai, China, November 2024.
- "Geo-AI Model Applications in Remote Sensing Data Based on Workflows." Presented at the Spatiotemporal Innovation Forum 2024, Shanghai, China, August 2024.
- "Dynamic Evolution of Carbon Storage in China's Terrestrial Ecosystems from 1990 to 2022." Presented at the 2024 Annual Meeting of the American Association of Geographers (AAG), Honolulu, HI, USA, April 2024.
- "Forest Change and Its Carbon Storage in China from 1990 to 2021." Presented at the Symposium on Spatiotemporal Data Science, Harvard University, Cambridge, MA, USA, July 2023.
- "Forest Change and Its Litter Carbon Storage in China from 1990 to 2021." Presented at the 2023 Annual Meeting of the American Association of Geographers (AAG), Denver, CO, USA, March 2023.
- "A Comparative Analysis of PSR and OHI Evaluation Methods for Shanghai Ocean Ecosystem Health." Presented at the China National Conference in Geography, Xi'an, China, August 2018.
- "Urban Green Space Change in Shanghai." Presented at the Global Land Programme 3rd Open Science Meeting, October 2016.

REFEREES

Xianjin Huang, Dr. & Professor, Nanjing University

Ruishan Chen, Dr. & Professor, Shanghai Jiao Tong University